

Public RNA-seq data
(NCBI SRA)

RNA-seq Read Processing, Mapping,
Counting and Normalization
(Nextflow)

Kruskal's minimum spanning tree

Rank (DEG Study 1) Rank (DEG Study 2) Rank (DEG Study 3)

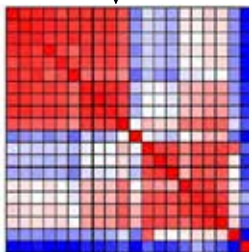
Calculate Spearman Correlation
 $\text{cor}(2,3) > \text{cor}(1,2) > \text{cor}(1,3)$

Rank(Rank 2 + Rank 3) + Rank 1 Borda Merge

Combined Rank of Genes for Pathogen #1

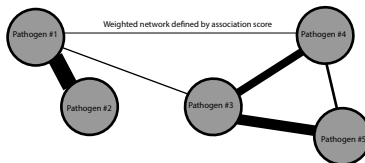
Pathogen #1
Pathogen #2
Pathogen #3
Pathogen #4
Pathogen #5

Pair-wise GSEA association analysis
(Kolmogorov-Smirnov Stats)



Pair-wise
Association Score (AS)

Pathogen #1
Pathogen #2
Pathogen #3
...



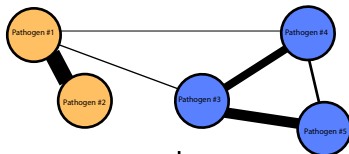
Community Detection / Clustering

Parameter-free affinity Propagation

Modularity Maximization based on attributes

Stochastic Block Model

Metadata-guided SBM (neoSBM)



Community-independent attributes?

Common pathogen attributes within community?

Common host response across community?

Community-specific host response?

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